HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • **Stamp 1G — Full-Time Work Eligible**

PROFESSIONAL EXPERIENCE — 3.5 YEARS

Software Engineer — HCL Technologies, Chennai, India Sep 2021 – Jan 2025

- Developed and deployed **Python** backend services for IoT device communication and data ingestion on **AWS Lambda**, handling approximately **4,500 events per hour**.
- Designed and implemented event-driven architectures using **AWS CDK** to provision and manage cloud infrastructure for the KitchenOS platform, ensuring scalable and resilient backend operations.
- Wrote **Python** scripts to automate ETL processes for device telemetry data, transforming raw inputs into structured formats for analysis and downstream processing, processing up to **50 GB of data daily**.
- Collaborated with cross-functional teams to define API contracts and integrate backend services with mobile clients and third-party systems, ensuring seamless data flow and functionality for kitchen appliances.
- Implemented robust logging and monitoring solutions using **AWS CloudWatch**, enabling proactive identification and resolution of production issues, reducing P1 incidents by **15%**.
- Containerized backend services using **Docker** for consistent deployment across development, staging, and production environments, simplifying the CI/CD pipeline managed by **Jenkins**.
- Participated in code reviews for a team of 4 engineers, providing constructive feedback to maintain code quality and adhere to best practices for Python backend development.

SKILLS

Languages – Python, Java, SQL, R

Cloud Platforms – AWS (Lambda, CDK, EC2, S3, CloudWatch, IAM)

Backend Frameworks & Tools – FastAPI, Flask, Spring Boot, Docker, Jenkins, Git

Data & Analytics – pandas, NumPy, PostgreSQL, MySQL, Oracle, Power BI, DAX, Tableau

IoT & Event-Driven – Event-Driven Architecture, IoT Devices, Message Queues, API Design

PROJECTS

IoT Device Communication Platform (HCL Technologies)

- Developed a **Python** backend service using **Flask** to manage communication between IoT kitchen appliances and the central platform, processing over **10,000 device commands** weekly.
- Implemented secure communication protocols for device authentication and data transmission, ensuring the integrity and privacy of sensitive operational data.
- Integrated with **AWS S3** for efficient storage of device logs and telemetry data, facilitating historical analysis and debugging.

Smart Kitchen Appliance Data Ingestion Pipeline

- Engineered a scalable data ingestion pipeline using **Python** and **AWS Lambda** to process real-time data streams from smart kitchen devices, handling approximately **3,000 messages per minute**.
- Designed and implemented data transformation logic using **pandas** to structure raw JSON payloads into a queryable format for analytics dashboards.
- Wrote unit tests with **pytest** to ensure the reliability and accuracy of data processing steps, achieving over **90% code coverage**.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India 2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)